

Artificial Intelligence & RAG Architecture Whitepaper

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1. Executive Summary

Retrieval-Augmented Generation (RAG) bridges the gap between static LLM memory and dynamic external data.

By retrieving authoritative context prior to generating answers, hallucination is minimized and factual precision is maximized.

This system provides enterprise teams with verified citations and traceability back to source pages.

2. Technical Benchmarks & Vector Databases

The system utilizes Qdrant as the vector database, paired with BGE-small-en embeddings (384 dimensions). Under benchmark testing with 50,000 document chunks:

- Retrieval Latency: 12 milliseconds (p95)
- Question Answering Accuracy: 96.4%
- Cosine similarity threshold for evidence validation: 0.35
- Chunking configuration: 700 characters with 150 character sliding overlap.